

VINUL GAJAMAN

Software Engineer | Full-Stack & Mobile

puldithvinul@gmail.com | +94 71 033 4443 | Colombo, Sri Lanka

linkedin.com/in/vinul-gajaman-a788a92b8 | github.com/VinulGajaman

Professional Summary

Software Engineer with 2+ years of experience building full-stack, mobile, and real-time systems across enterprise, IoT, and Web3 domains. Experienced in NestJS, Spring Boot, React/Next.js, React Native, and Kotlin/Jetpack Compose, with production work in REST APIs, microservices, Docker/Kubernetes, and CI/CD. Recently built a real-time RFID access-control platform for airport security and non-custodial crypto wallets supporting Bitcoin and 7 EVM chains.

Technical Skills

Programming Languages:
Java, TypeScript, JavaScript, Kotlin, SQL

Databases:
PostgreSQL, MySQL, TimescaleDB, MongoDB, SQLite/PowerSync, Firebase

Generative AI:
OpenAI API, Gemini API, RAG, Claude Code

Backend:
NestJS, Spring Boot, Spring AI, REST APIs, Microservices, Socket.io, JWT/Passport

Web3:
ethers.js, Web3j, BitcoinJ, BDK, BIP39/32/44, 1inch API

Frontend & Mobile:
React.js, Next.js, React Native (Expo), Jetpack Compose, Tailwind CSS, Material-UI

Cloud & DevOps:
Docker, Kubernetes, Azure AKS, GitHub Actions, Git, Bitbucket, JIRA

Professional Experience

11/2025 – Present
Colombo, Sri Lanka

Coding Legends (Pvt) Ltd, *Junior Software Engineer*

- Engineered a real-time RFID and biometric access-control platform for Airport and Aviation Sri Lanka (NestJS, Next.js, Socket.io, TimescaleDB), cutting scan-to-display latency from ~2s to ~50–200ms with an in-memory O(1) authorization engine.
- Owned the platform's RFID backend over persistent TCP connections to UHF readers (binary protocol), with in/out movement detection, fail-closed RBAC, and a 3-tier TimescaleDB data lifecycle.
- Developed a native Android app (Kotlin, Jetpack Compose) for handheld UHF RFID scanners, integrating the UHFRManager hardware SDK and ML Kit QR scanning for real-time ticket and member verification.
- Built non-custodial multi-chain crypto wallets for Android (Kotlin) and iOS/Android (React Native) supporting Bitcoin and 7 EVM chains, with on-device key derivation, AES-GCM seed encryption via Android Keystore, and biometric/PIN security.
- Designed and built NestJS wallet backends, including JWT/Passport authentication built from scratch, admin token and market management, a 1inch-powered swap service, staking APIs, and unsigned-transaction building for client-side signing.
- Built an offline-first construction expense platform (React Native, NestJS, PostgreSQL, PowerSync) with instant offline writes and maker-checker approvals, and diagnosed and fixed a head-of-line blocking bug in the photo-upload queue, cutting upload time from ~20 minutes to seconds.

06/2024 – 11/2025
Colombo, Sri Lanka

DMS Software Technologies (Pvt) Ltd, *Software Engineer Intern*

Tour Management System - full-stack travel platform (Spring Boot · React.js · Docker · MySQL, microservices)

- Designed and optimized REST API endpoints, reducing average response time by 35% through efficient query design and data handling.
- Built 15 responsive UI components using Material-UI, enabling advanced filtering and pagination for datasets of 5,000+ records.
- Developed and deployed an AI-powered chatbot using OpenAI API and RAG architecture, handling 100+ daily queries and reducing manual support tickets by 50%.
- Containerized enterprise applications using Docker, reducing setup time from hours to under 5 minutes across development, QA, and production environments.
- Led data migration for two legacy systems with 100,000+ records using MySQL Workbench, completing the project ahead of schedule with zero data loss.
- Maintained five production environments supporting 150+ concurrent users, delivering end-to-end technical support and resolving critical production issues.

Projects

AASL Airport Access Control Platform | Coding Legends
Tech: NestJS, Next.js, Socket.io, TimescaleDB
Real-time access-control platform for airports, combining UHF RFID tracking and facial recognition to authorize personnel movement against an external permit authority. Built the RFID backend in NestJS with persistent TCP connections to UHF readers over a binary protocol, a Socket.io gateway cutting scan-to-display latency from ~2s to ~50–200ms, and an in-memory O(1) authorization engine with full restart durability. Also built movement-detection logic (antenna pairings, in/out direction), a 3-tier TimescaleDB data lifecycle, streaming CSV exports, and fail-closed RBAC, and rebuilt the live monitoring console in Next.js.

Android UHF RFID Handheld Scanner | Coding Legends
Tech: Kotlin, Jetpack Compose, Retrofit, ML Kit
Native Android app for handheld UHF RFID devices, built for AASL. Interfaces with the UHFRManager hardware SDK via serial port to read tag EPC data in real time, uploads scans to a REST API, and verifies tickets and members through ML Kit QR scanning. Includes JWT authentication with token refresh and configurable backend URLs.

Offline-First Construction Expense Platform | Coding Legends
Tech: React Native (Expo), NestJS, PostgreSQL, PowerSync, SQLite
Local-first expense and bill-tracking app for construction teams without reliable coverage. Every write commits instantly to local SQLite and syncs on reconnect via PowerSync, with a server-side approval state machine enforcing maker-checker controls on every money-moving action. Fixed a head-of-line blocking bug in the photo upload queue, cutting upload time from ~20 minutes to seconds.

TechSalary LK (Cloud-Native Salary Transparency Platform)
Tech: Docker, Kubernetes (Azure AKS), GitHub Actions, Jest
Salary-transparency web app built as 7 containerized microservices on Azure Kubernetes Service, with a Backend-for-Frontend gateway handling JWT auth and routing. Implemented a community approval workflow (submissions go live after 3 upvotes) feeding search and aggregate statistics. Automated full CI/CD with GitHub Actions: parallel Jest tests, Docker builds to Azure Container Registry, and kubectl rolling updates for zero-downtime deploys.

AI Chatbot Backend | DMS Software Technologies
Tech: Spring AI, OpenAI API, RAG
Chatbot backend using Spring AI with Retrieval-Augmented Generation over a product PDF knowledge base, exposing REST endpoints for domain-specific AI support.

Jova Multi-Chain Crypto Wallet (Android + Backend) | Coding Legends
Tech: Kotlin, Jetpack Compose, Web3j, NestJS
*API*Non-custodial wallet for Bitcoin and 7 EVM chains. Android app with BIP39/32/44 key derivation, salted PIN with progressive lockout, root detection, and on-device EVM signing. Backend with CoinGecko token registry sync, cached price and OHLC chart pipelines, and staking APIs built from the contract ABI.

Education

2025 – Present
UK
MSc in Advanced Software Engineering (Expected 09/2027),
Informatics Institute of Technology (IIT), Colombo
Awarded by University of Westminster, UK | 2025 – Present

2020 – 2024
UK
BSc (Hons) in Software Engineering, *Java Institute for Advanced Technology, Sri Lanka*
Awarded by Birmingham City University, UK | 2020 – 2024

References

M M P M Gunarathna, *Chief Finance Manager, Finance Division,* Sri Lanka Ports Authority
pabasara@slpa.lk, 071-8688372

K.J. Lokugama, *Senior Manager, Information Systems Division,* Sri Lanka Ports Authority
kusal@slpa.lk, 071-5306434, 071-3257867